

Software Requirement Specification (SRS)

Project Title

Workflow solution for the ISO Document Library

Revision History

Version	Date	Author	Description of Changes
V1.0	08/06/2026	Anjalee	Initial SRS draft
V2.0	08/07/2026	Anjalee	Added ER Diagram, UI/UX wireframes, Business Logic; revised Use Case Diagram; corrected Flow Diagram
V3.0	08/10/2.06	Savindi & Anjalee	Final consolidation: merged functional requirements, hybrid duplicate-detection logic, unified data model and ER/use-case diagrams, and combined UI flow.

Contents

1.Introduction	3
1.1 Purpose.....	3
1.2 Scope.....	3
1.3 Intended audience	3
1.4 Definition.....	3
2.Overall Description	4
2.1 Product Perspective.....	4
2.2 Product Functions	4
2.3 User Classes and Characteristics.....	4
2.4 Operating Environment	5
2.5 Design and Implementation Constrains.....	5
2.7 Assumptions and Dependencies	5
2.8 User Interface	6
3. system Features and requirements	9
3.1 Functional Requirements	9
3.5 Non-functional Requirements	10
3.2 Entity relationship diagram.....	11
3.2 Use cases Diagram	12
3.3 External Interface Requirement	12
3.4 Database Requirement	13
4. Business Logic	14
4.1 Login / Authentication	14
4.2 Upload, Validation and Storage	14
4.3 Existing Document Check and Content Comparison (hybrid logic).....	14
4.4 Version Management - Version History	14
4.5 Approval Matrix Routing	15
4.6 Approval Decision.....	15
4.7 Resubmission.....	15
4.8 Notification and Audit	15
4.9 Role-Based Access Control.....	16

1.Introduction

1.1 Purpose

The purpose of this Software Requirement specification is to define the requirements for developing an ISO Document Library Workflow solution for the GPSL. System will manage ISO Documents by allowing the users to upload documents to the SharePoint, Validate the document format, Compare the document version, route the documents through an approval workflow, replacing the current manual, email-based process. As the final step Securely store the approved document and publishing the SharePoint.

1.2 Scope

The proposed system shall allow GPSL employees to:

- Upload documents for review through a web application.
- Have documents validated against the GPSL ISO template and naming convention.
- Have the system detect whether a document already exists and compare it against the current version.
- Have version differences (added, removed, or modified sections) identified automatically.
- Have documents routed to the correct approver via a configurable approval matrix.
- Be notified of approval or rejection outcomes and resubmit corrected documents.
- Have approved documents published to Microsoft SharePoint as the controlled current version.

1.3 Intended audience

- GPSL Employees (Document Creators)
- Approvers / Reviewers (Department Managers)
- GPSL Management
- System Administrator

1.4 Definition

- ISO - International Organization for Standardization
- SharePoint- Microsoft Document management Platform
- Approver - Person who is responsible for approving documents.
- Approval matrix - A mapping of Document Type and Department to a Designated Approver.

- Content Comparison -The process of determining whether an uploaded document is identical to, or a modified version of, an existing document.

2.Overall Description

2.1 Product Perspective

Workflow solution will integrate with GPSL's existing Microsoft SharePoint environment and company email/authentication systems. It does not replace SharePoint storage; it adds a validation, comparison, and approval-routing layer in front of it, so that only ISO-compliant, approved documents become the published controlled version.

2.2 Product Functions

The system Shall;

- Authenticate users via their GPSL company account.
- Accept document uploads and validate format, naming, and mandatory metadata.
- Detect duplicate or existing documents and compare content against the current version.
- Identify added, removed, or modified sections between versions.
- Automatically create and retain new document versions.
- Determine the correct approver via the approval matrix and route the document.
- Record approval or rejection decisions, with mandatory comments on rejection.
- Notify users of all status changes and support resubmission of corrected documents.
- Publish approved documents to SharePoint and maintain a full audit trail.

2.3 User Classes and Characteristics

- **Employee (Document Creator)**- Uploads documents and views the status/history of their own submissions only.
- **Approver / Reviewer** - Departmental staff with basic-to-moderate computer literacy, familiar with the document types relevant to their department. Reviews, approves, or rejects documents routed to them; cannot see documents assigned to other approvers.
- **Administrator** - IT or document-control staff with moderate technical proficiency. Maintains the approval matrix, users and roles, ISO templates and validation rules, and generates reports. Does not hold approval rights unless separately assigned as an Approver.

2.4 Operating Environment

The system will operate as a web-based application accessible through standard company web browsers over the GPSL company network. It shall be integrated with Microsoft SharePoint (Online or on premises, as deployed at GPSL) for document storage and with the company Email Server for notifications. No client-side installation shall be required beyond a web browser.

2.5 Design and Implementation Constrain

- Documents must conform to the GPSL ISO template (heading structure, metadata fields, numbering); non-conforming uploads are rejected at upload with a clear error message.
- The approver is always determined programmatically from the approval matrix, never selected manually by the uploader.
- Only one active/published version of a given document may exist in SharePoint at a time; new versions supersede prior versions in a traceable manner — prior versions are retained, not overwritten.
- The system must integrate with GPSL's existing SharePoint environment and use GPSL's existing company account system for authentication rather than a separate login.
- The approval matrix must be configurable by an Administrator without code changes.

2.7 Assumptions and Dependencies

- Assumes all users have valid, active GPSL company accounts prior to using the system.
- Assumes GPSL's SharePoint instance and Email Server are always available and accessible to the system.
- Assuming the approval matrix is kept up to date by an Administrator, and one approver (or defined chain) is configured per document type/department combination.
- Assumes uploaded documents are in supported formats (to be defined in validation rules) and within any agreed file size limits.

2.8 User Interface

Login Page



ISO Document Library

SIGN IN WITH YOUR GPSL ACCOUNT

Company Email

Password

SIGN IN

Authenticated via GPSL company account

Document uploading page

ISO document libraryDashboardUploadDocument historyEmployeeAS

Upload document

Upload a new or updated ISO document for review



Drag file here or click to browse
PDF or DOCX, max 25MB

Document title

Document type

Department

Version

Cancel

Upload Document

GPSL ISO document library · v1.0

Support

Approval Page

ISO document library

Dashboard

Upload

Document history

Employee

AS

Review document

ISO 9001 quality manual · v2 · quality assurance

Uploaded files

Uploaded by A. Silva · 2 hours ago

Find Modified Section

View Details

Comments

Reject

Approve

GPSSL ISO document library · v1.0

Support

User Dashboard

ISO document library

Dashboard

Upload

Document history

Employee

AS

Pending review

6

Approved this month

23

Rejected

3

My documents

Search by id or name.....

Upload New ↗

Document_ID	Document	Upload_date	Version	Status	Approval_date
003	Document name 1	01/08/2026	V1	Pending	pending
002	Document name 2	31/07/2026	V2	Approved	01/08/2026
001	Document name 3	30/07/2026	V3	Rejected	01/08/2026

GPSSL ISO document library · v1.0

Support

Popup View

My

Search

Document Details

Document ID

Document

DOC

xyz

Upload Date

Approval Date

Approval by

Department

Sharepoint link

Status

Description

Comments

Download

3. system Features and requirements

3.1 Functional Requirements

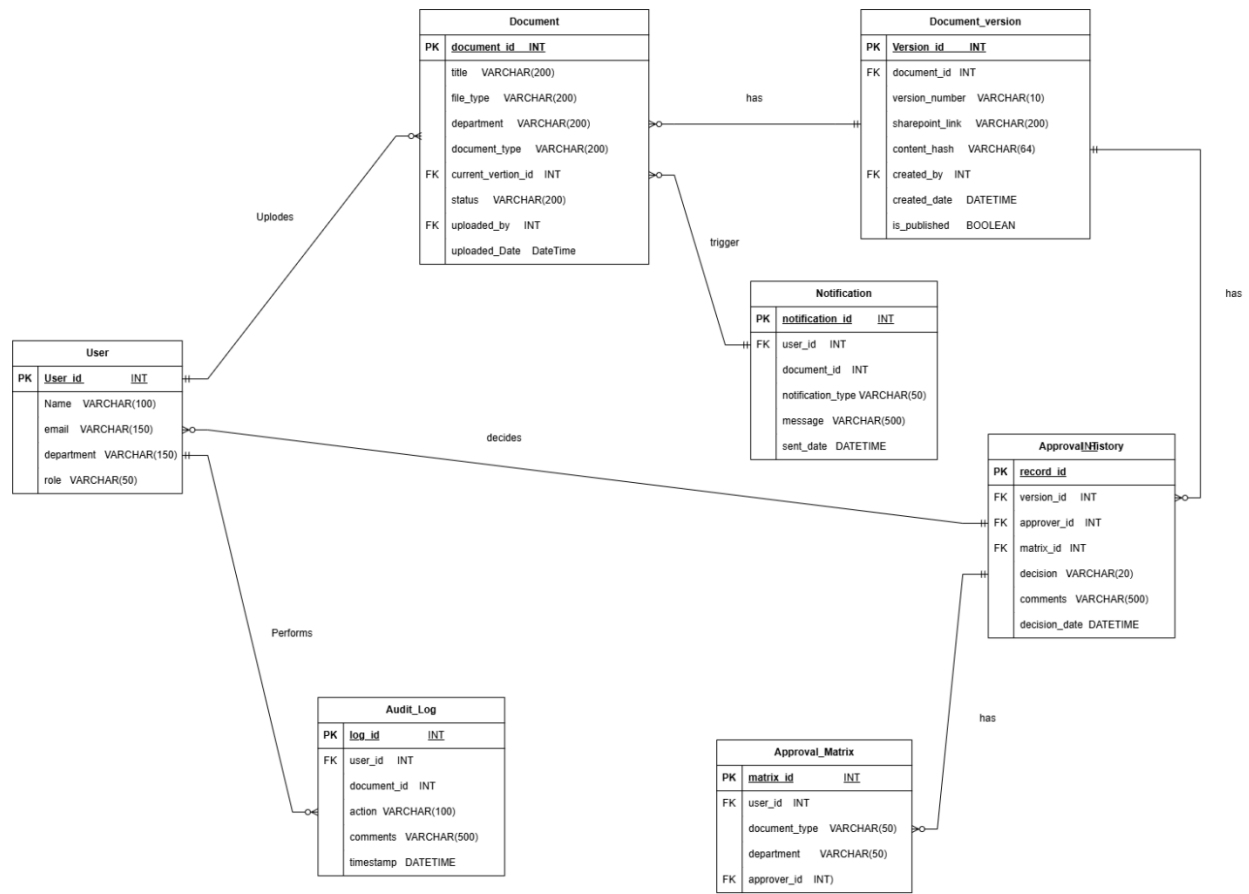
- **FR-01 User Login** - The system shall authenticate users via their GPSL company account (single sign-on), no separate password is stored. After a defined number of failed attempts, the standard company lockout policy applies. Sessions time out after a period of inactivity (e.g., 30 minutes).
- **FR-02 Upload Document** - Employees shall be able to upload a document together with Title, Document Type, Department.
- **FR-03 Validate Document** - The system shall verify file format (PDF/DOCX, configurable), the naming convention [DocumentType]-[Department].ext, and mandatory metadata fields. On failure, the upload is rejected with a specific reason and the user is returned to the upload screen; no document record is created.
- **FR-04 Existing Document Check** - The system shall search for a metadata match (Title + Document Type + Department). If found, the document proceeds to Content Comparison against that specific document; if not found, it still proceeds to Content Comparison against stored documents generally, since identical content could exist under a different name.
- **FR-05 Content Comparison (hybrid)** - The system shall first generate a content hash/checksum of the uploaded file and compare it to the relevant existing version(s). If identical, the upload is rejected as a duplicate, and the user is notified with a reference to the existing document. If the metadata matches but the hash differs, the system shall perform a section-by-section comparison to classify differences as added, removed, or modified sections (optionally AI-assisted) before accepting the document as a new version. Partial textual similarity to an unrelated document, no metadata match and no exact hash match, shall never block an upload; only an exact hash match is treated as a duplicate.
- **FR-06 Version Management** - When differences are confirmed, the system shall create a new Document_Version record (incrementing the version number) and update the parent Document's current_version_id to point to it. Every prior Document_Version row is retained permanently and is never deleted or overwritten - this row-per-version table is the authoritative version history, viewable in full from the Document History screen.
- **FR-07 Approval Matrix Routing** - The system shall determine the approver by looking up the approval matrix on Document Type and Department. If no match is found, the document is flagged to the Administrator instead of failing silently. Each Document Type + Department + Approver combination maps to exactly one designated approver.
- **FR-08 Approval Decision** - Approvers may Approve or Reject a document. Approve sets status to "Approved" and stores the final version in SharePoint. Reject sets status to "Rejected", requires a mandatory comment, and notifies the uploader with that comment.

- **FR-09 Notification** - The system shall notify the relevant user on every status change: validation failure, duplicate rejection, approval, or rejection.
- **FR-10 Resubmission** - A rejected document returns to the uploader in an editable state with the approver's comments attached. On resubmission it re-enters the workflow at Validation (FR-03), not directly at Approval, and receives a new incremented version number.
- **FR-11 Publish Document** - Only an approved version shall be published to SharePoint, stored under a structured path such as /ISO-Library/[Department]/[Document Type]/.
- **FR-12 Audit Trail** - Every workflow action - upload, validation result, comparison result, routing, and approval decision - shall be written to the audit log, capturing User, Action, Document, Timestamp, and any comments.

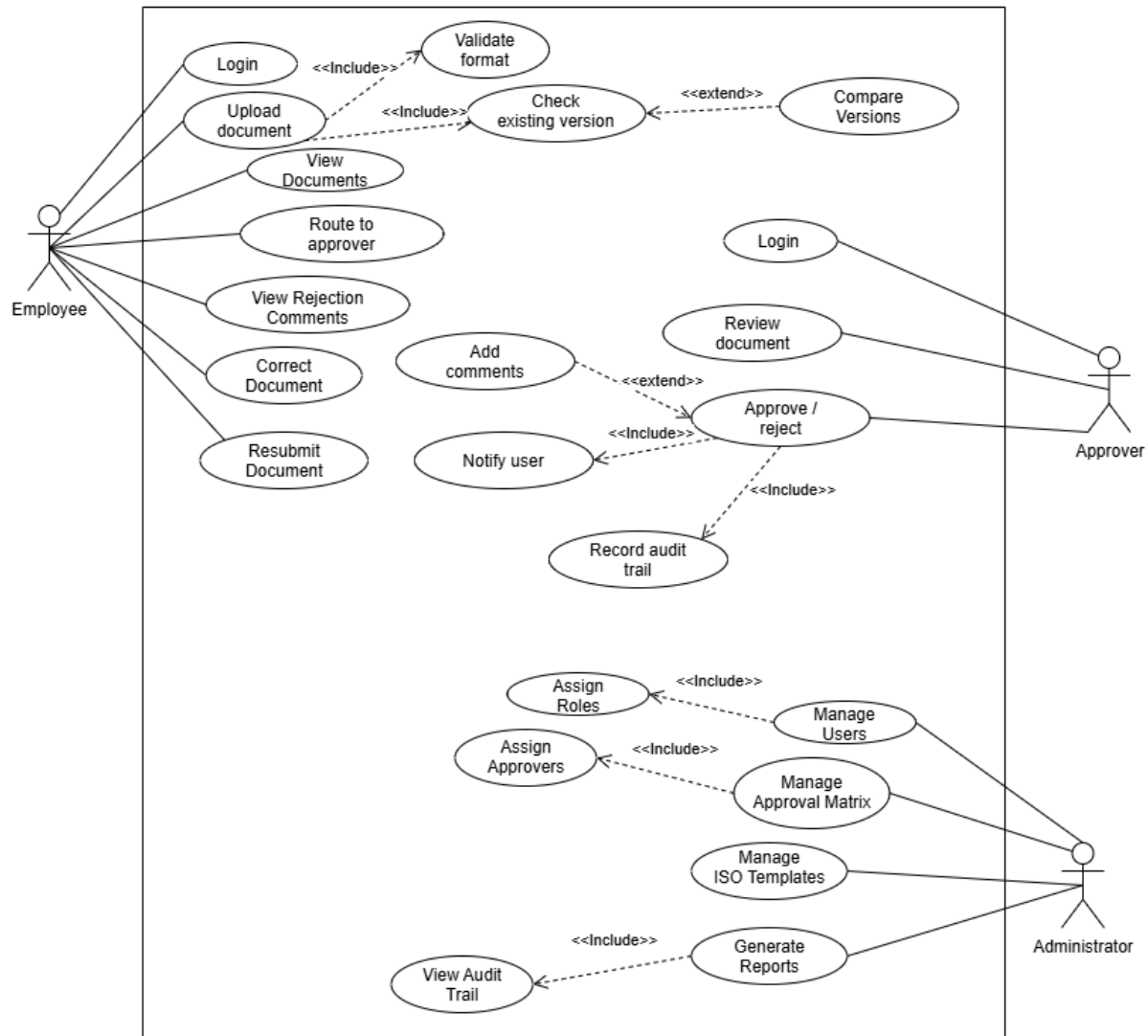
3.5 Non-functional Requirements

- **Performance** - Document validation shall be completed within 5 seconds for a normalized document; the system shall support at least 100 concurrent users.
- **Security** - Role-based access control; only authorized, authenticated users may access the system.
- **Reliability** - 99.9% system availability (maximum allowed downtime of approximately 8 hours 48 minutes per year).
- **Usability** - A simple, easy-to-navigate interface with clear, specific error messages.
- **Scalability** - The system shall support a growing number of users and documents without redesign.
- **Maintainability** - The approval matrix and validation rules shall be configurable by an Administrator without developer involvement.

3.2 Entity relationship diagram



3.2 Use cases Diagram



3.3 External Interface Requirement

- **User Interface:** Login Page, Dashboard, Upload Screen, Approval Screen, Document History.
- **Hardware Interface:** User PC, Company Network
- **Software interface:** Microsoft SharePoint, Email Server

3.4 Database Requirement

The system database stores document metadata and workflow state; the documents themselves are stored in SharePoint.

Entity	Key Fields	Notes
User	user_id (PK, INT), name (VARCHAR100), email (VARCHAR150), department (VARCHAR50), role (VARCHAR50)	Role: Employee / Approver / Administrator
Document	document_id (PK, INT), title (VARCHAR200), file_type (VARCHAR10), department (VARCHAR50), document_type (VARCHAR50), current_version_id (FK, INT), status (VARCHAR20), uploaded_by (FK, INT), upload_date (DATETIME)	status reflects the current/latest version only
Document Version	version_id (PK, INT), document_id (FK, INT), version_number (VARCHAR10), sharepoint_link (VARCHAR250), content_hash (VARCHAR64), change_summary (VARCHAR500), created_by (FK, INT), created_date (DATETIME), is_published (BOOLEAN)	One row per version; prior rows are never deleted or overwritten - this is the version history
Approval Matrix	matrix_id (PK, INT), document_type (VARCHAR50), department (VARCHAR50), approver_id (FK, INT)	One approver per Document Type + Department
Approval History	record_id (PK, INT), version_id (FK, INT), approver_id (FK, INT), matrix_id (FK, INT), decision (VARCHAR20), comments (VARCHAR500), decision_date (DATETIME)	One decision per Document_Version, not per Document - supports resubmission history
Notification	notification_id (PK, INT), user_id (FK, INT), document_id (FK, INT), notification_type (VARCHAR50), message (VARCHAR500), sent_date (DATETIME)	Linked to both the recipient and the document
Audit Log	log_id (PK, INT), user_id (FK, INT), document_id (FK, INT), action (VARCHAR100), comments (VARCHAR500), timestamp (DATETIME)	Captures every workflow action end-to-end

4. Business Logic

4.1 Login / Authentication

- Authentication uses the existing GPSL company account; no separate password is created or stored.
- Failed logins show a generic error and allow retry; repeated failures trigger the standard company lockout policy.
- Sessions time out after a period of inactivity and require re-authentication.
- The user's role is retrieved from the company account system and determines available screens and actions.

4.2 Upload, Validation and Storage

- A document must match the GPSL ISO template and naming convention to be accepted; failures return the user to the upload step with a specific reason and no record is created.
- Approved documents are stored in SharePoint under a structured path based on Department and Document Type; previous versions are retained and remain accessible through Document History.

4.3 Existing Document Check and Content Comparison (hybrid logic)

- Step 1 - Metadata match: search by Title + Document Type + Department. A match marks the upload as a candidate new version; no match still proceeds to comparison against all stored documents, since identical content may exist under a different name.
- Step 2 - Hash check: compute a checksum of the uploaded file against the relevant existing version(s). An identical hash means the upload is a duplicate - reject it, notify the user, and reference the existing document; stop the workflow (no new version, no approval triggered).
- Step 3 - Section-level diff: if the hash differs, compare the two documents section by section (not a byte diff) and classify differences as added, removed, or modified sections. This gives the approver a precise view of what changed, and can be surfaced via an AI-assisted “find modified sections” aid on the approval screen.
- If no metadata match and no content match exist anywhere, the document is treated as genuinely new and proceeds directly to approval routing.

4.4 Version Management - Version History

- Confirmed differences create a new Document_Version row (document_id, version_number, sharepoint_link, content_hash, created_by, created_date) rather than modifying the existing one; the Document record's current_version_id is then updated to point at it.

- Every Document_Version row created is kept permanently, giving each document a full, ordered version history - including rejected and superseded versions - that is never deleted.
- The Document History screen (§6) reads directly from Document_Version, so a user or approver can open any past version, see who created it and when, and see which decision (Approval_History) applied to it.
- Only the version currently marked is_published = true in SharePoint is the live controlled copy; all other rows remain retrievable for audit but are not the active version.

4.5 Approval Matrix Routing

- The approver is looked up from the approval matrix using Document Type and Department.
- If no configured approver is found, the document is flagged to the Administrator rather than failing silently.
- Only one designated approver is supported per Document Type + Department combination; this is treated as a system constraint, not a routing choice.

4.6 Approval Decision

- Approve: status becomes “Approved” and the final version is published to SharePoint.
- Reject: status becomes “Rejected”; a comment is mandatory; the uploader is notified with the approver's comments. The last-approved published version, if any, remains the live version; the rejected version is retained for audit but never published.

4.7 Resubmission

- A rejected document returns to the uploader in an editable state with the approver's comments attached.
- Resubmission re-enters the workflow at Validation, not directly at Approval, since corrections may affect format or naming.
- Each resubmission increments the version number so the new attempt never collides with the rejected version.

4.8 Notification and Audit

- A notification is triggered on every status change: validation failed, duplicate rejected, approved, or rejected.
- Every workflow action - upload, validation result, comparison result, routing, and approval decision - is written to the audit log with User, Action, Document, Timestamp, and any comments.

4.9 Role-Based Access Control

- Employee: may upload documents and view the status of only their own submissions.
- Approver: may view, approve, or reject only documents routed to them; cannot see documents assigned to other approvers.
- Administrator: full access to manage the approval matrix, users, templates and validation rules, and to generate reports; no approval rights unless separately assigned as an Approver.